

## **Catalonia**

### **Catalan Competition Authority**

The Catalan Competition Authority (ACCO) initiated the ERICCA project in 2022 with the objectives of supporting sanctioning proceedings related to Catalan public tenders and proactively identifying tenders with a higher likelihood of anti-competitive agreements through artificial intelligence.

A significant initial challenge, obtaining quality data, was addressed through collaboration with the Department of Economy of the Generalitat of Catalonia, granting ACCO access to the Public Contracts Registry database. Additionally, other data sources are also used, such as those available through the Open Data Portal.

The ERICCA tool employs an unsupervised machine learning model that classifies companies into clusters based on their participation in tenders and calculates statistical parameters to detect potential collusion. Since mid-2023, this model has integrated international cartel data, transitioning into a supervised approach using a deep neural network.

Previous advancements during the period 2023-2024 laid crucial groundwork for the project's further development. Key improvements included the expansion of analyzed data to encompass low-value or 'minor contracts,' enhancing the detection of contract fragmentation. Additionally, significant upgrades were introduced in data visualization and reporting tools to facilitate the investigative process. The robustness of analyses was strengthened through the incorporation of advanced statistical parameters, including expected discount calculations and random forest methods.

Importantly, the supervised neural network model, trained on international datasets from Switzerland, Italy, and Japan, was successfully integrated, substantially enhancing the accuracy of identifying collusive patterns.

### **I. Main Advances During 2024-2025**

The developments of the ERICCA project during the years 2024-2025 further built on this foundation, introducing critical enhancements aimed at refining detection accuracy, broadening the application scope, and automating analysis processes. These improvements underline ACCO's ongoing commitment to

leveraging cutting-edge artificial intelligence techniques in ensuring fair competition in public procurement.

During 2025, significant improvements have been implemented in the application of the algorithm that have led to the direct identification of several cases of anti-competitive conduct in recent months. These improvements have emerged from the realization that the identification of anti-competitive conduct is more granular than initially thought.

The main discovery has been that markets are much more specific than what is usually found under economic scope variables such as CPV (Common Procurement Vocabulary). This reality has required a more intelligent delimitation of the potential ‘group’ of contracts on which to apply the algorithm.

The new methodology allows:

- More precise market segmentation: Identification of submarkets within apparently homogeneous CPV categories, taking into account factors such as geographical location, technical complexity of the service or product, and specific characteristics of the contracting body.
- Selective algorithm application: Instead of applying the algorithm uniformly to all tenders in a category, it now focuses on groups of contracts with similar characteristics that allow for more meaningful comparison.
- Detection of more subtle patterns: The improved granularity has allowed illegal behaviors to surface that were previously diluted in the analysis of overly broad categories.

This methodological evolution has proven to be key in bringing these illegal behaviors to light and has validated the effectiveness of the ERICCA tool as a proactive instrument for detecting anti-competitive practices.

## **II. Complementary Tools for the Detection of Collusive Practices**

### *A. Artificial Intelligence Agent for Document Review*

One of the most significant ongoing innovations during this period is the development of an artificial intelligence agent specialized in the automated review of technical specifications and official documents related to public procurement awards. This agent represents a crucial step forward in automating the supervision process and is expected to become a key tool for enhancing the accuracy of the ERICCA system.

The main functionalities currently being implemented in this agent include:

- **Automatic analysis of specifications:** The agent is designed to automatically process and analyze both technical and administrative specifications, with the aim of identifying elements that may indicate irregularities or conditions that could favor specific companies.
- **Review of award documents:** The system is configured to examine official award documentation, including minutes of contracting committees and technical reports, in order to detect inconsistencies or suspicious patterns.
- **Reduction of false positives:** One of the most valuable anticipated functionalities is the agent's capacity to reduce the number of false positives generated by the neural network-based AI model.
- Based on the analysis of bids submitted by companies, ERICCA can **identify procurement procedures** characterized by the absence or a very limited degree of competition. However, in some cases, this lack of competition is the result of restrictions introduced by the contracting authorities themselves through clauses governing the tender process (such as limiting the award of more than one lot to the same company). The contextual analysis of the documentation enables the agent to determine whether the reduced level of competition detected by ERICCA is attributable to the behavior of the participating companies or to the conditions established in the administrative specifications.
- **Cross-validation of results:** The agent is intended to serve as an additional layer of validation, contrasting the outputs of the neural network model with the information contained in official documentation, thereby providing a more accurate and reliable assessment.

This innovation has significantly reduced the number of false positives, improving the efficiency of the investigation process and allowing ACCO technicians to focus on cases with a real probability of anti-competitive conduct.

#### *B. Other Complementary Tools for the Detection of Collusive Practices*

Furthermore, lightweight software tools have been developed to assist in the detection of collusive practices. For example, one such tool automatically extracts metadata from electronic documents and enables verification of whether their authors are employees of competing firms.